titanic-eda-starter-2

April 14, 2025

1 Titanic Dataset - Exploratory Data Analysis (EDA)

Welcome! This notebook will guide you through the Titanic dataset EDA step by step.

[44]:
```python
#step: 1
# Import all the required libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np

# Optional for inline plotting in Jupyter
%matplotlib inline
```

[46]:
```python
#step: 2

train_df = pd.read_csv("train.csv")
```

[32]:
```python
# Step 3: Basic Info
train_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
```

```
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

[34]:
```
# Step 3: Summary Statistics
train_df.describe()
```

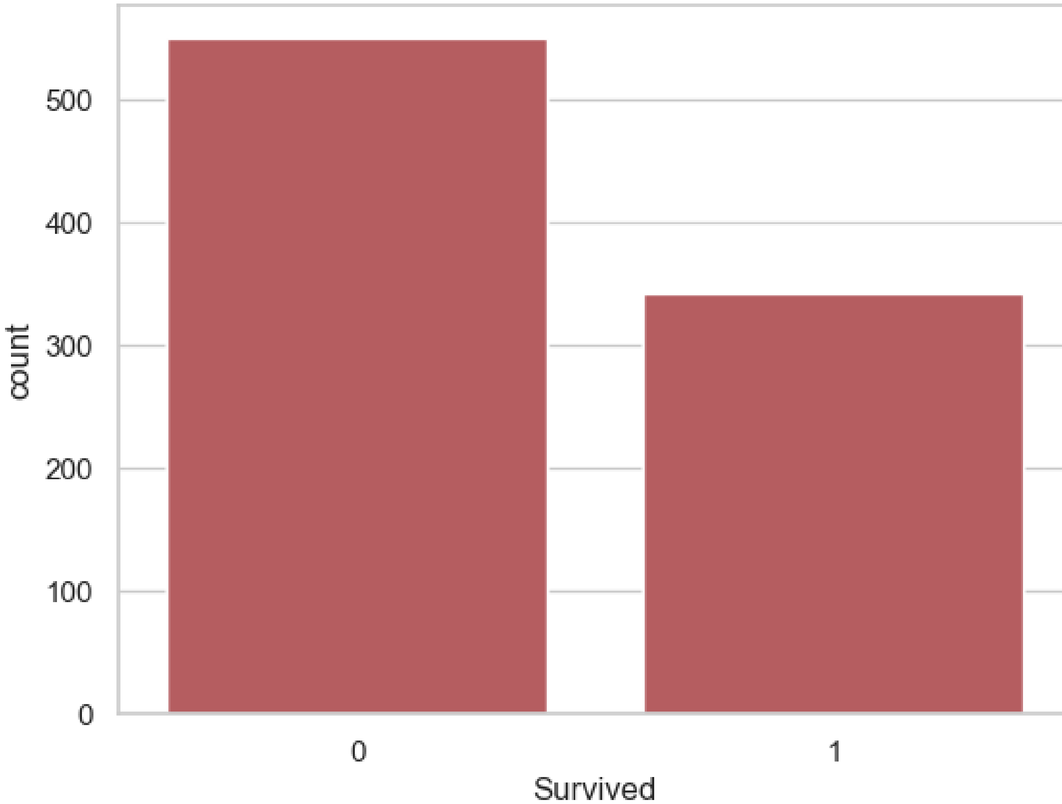
[34]:
```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

[36]:
```
# Step 3: Check Missing Values
train_df.isnull().sum()
```

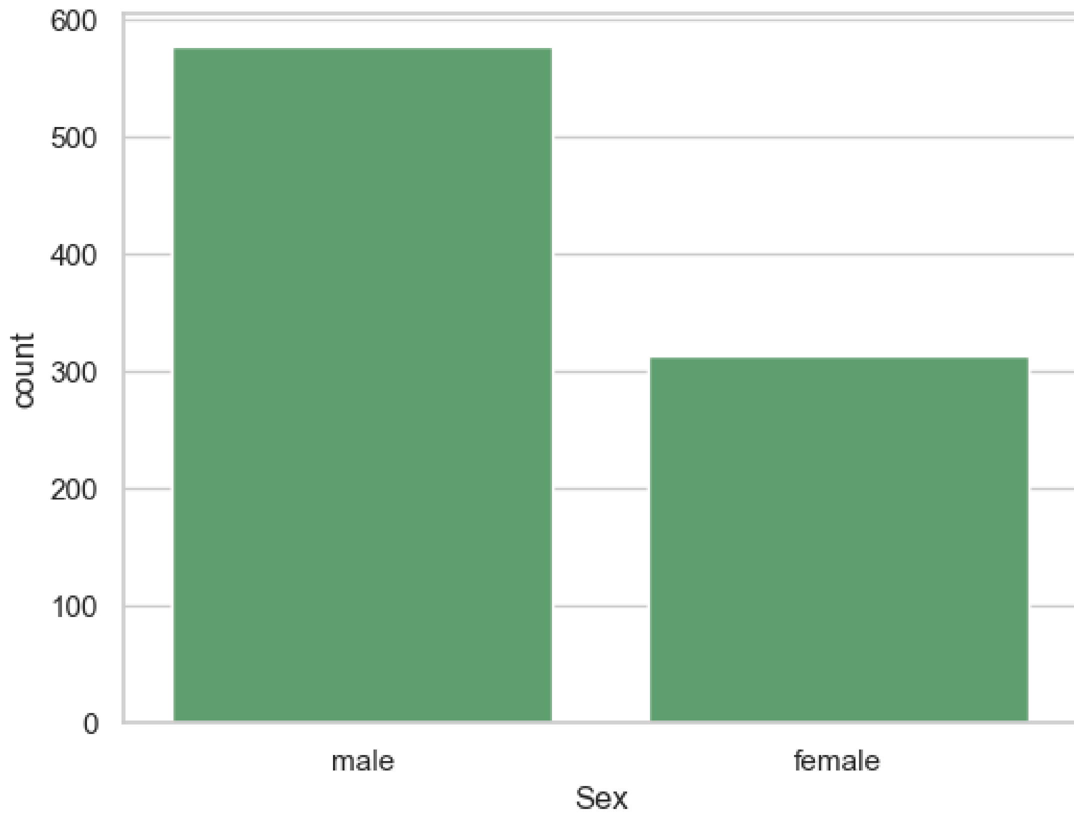
[36]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

1.0.1 STEP 4: Univariate Analysis (One Variable at a Time)

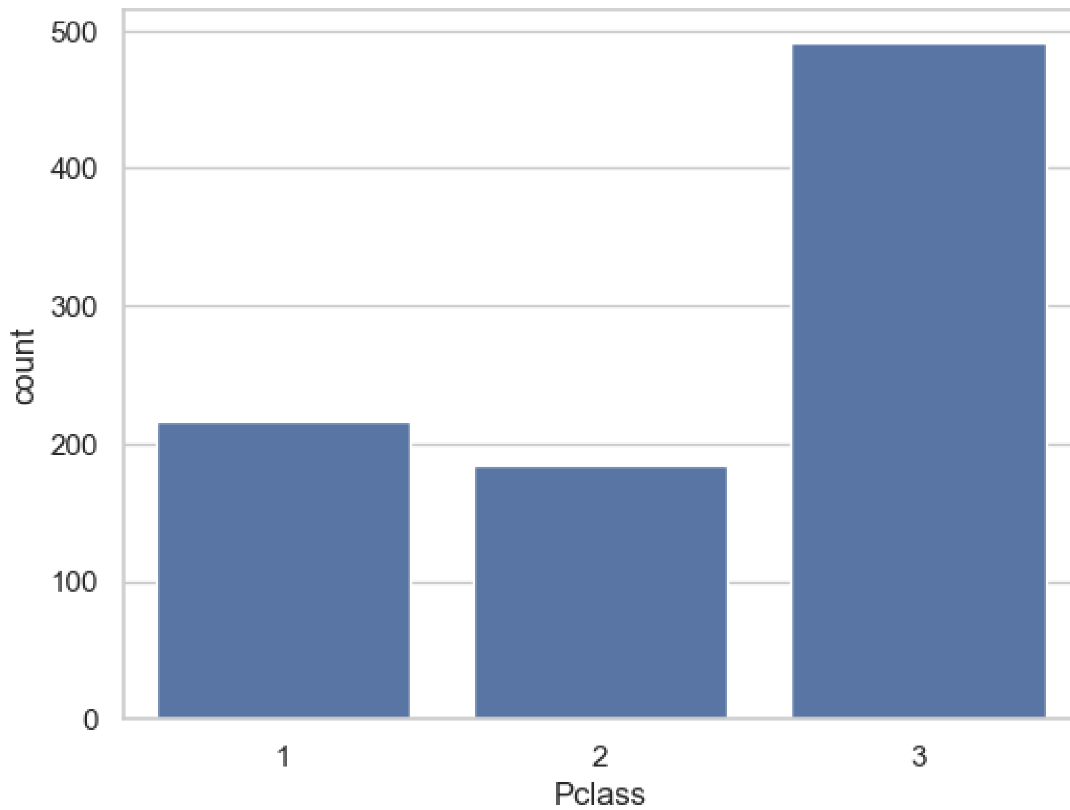
[52]:
```
#Target variable - Survival count:
import matplotlib.pyplot as plt  # Make sure matplotlib is imported!

sns.countplot(x='Survived', data=train_df)
plt.show()
```



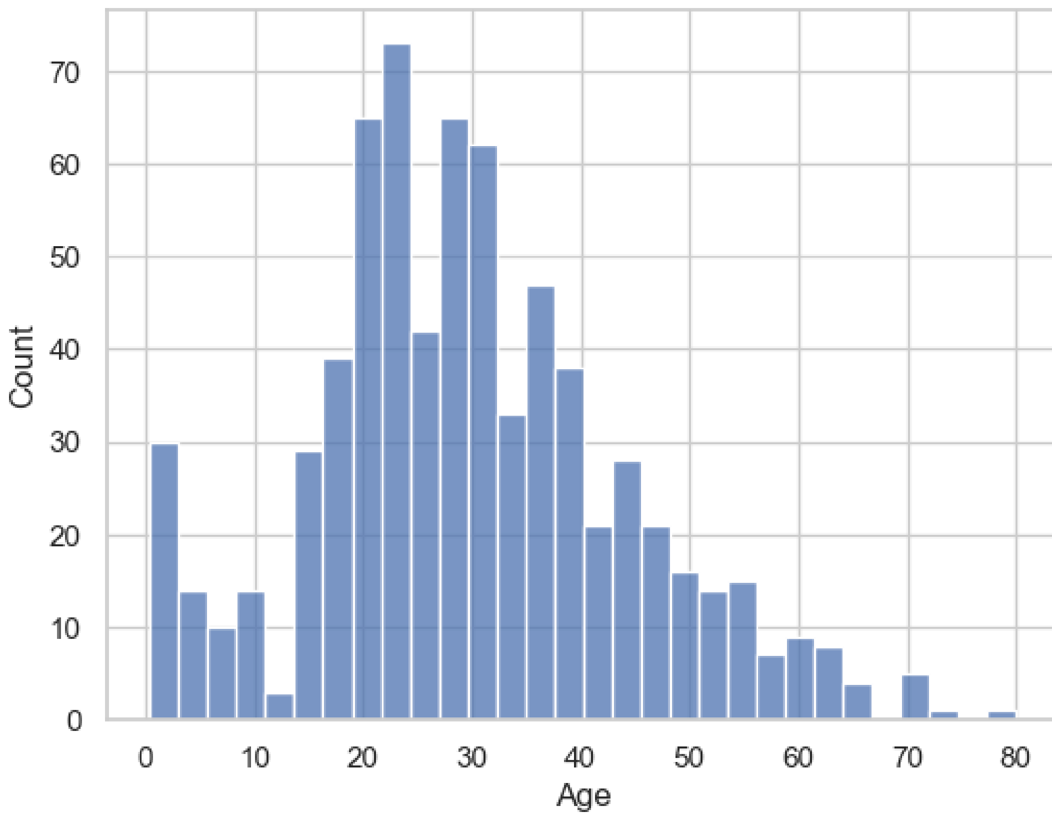
[58]:
```
#Gender distribution:
import matplotlib.pyplot as plt
sns.countplot(x='Sex', data=train_df)
plt.show()
```



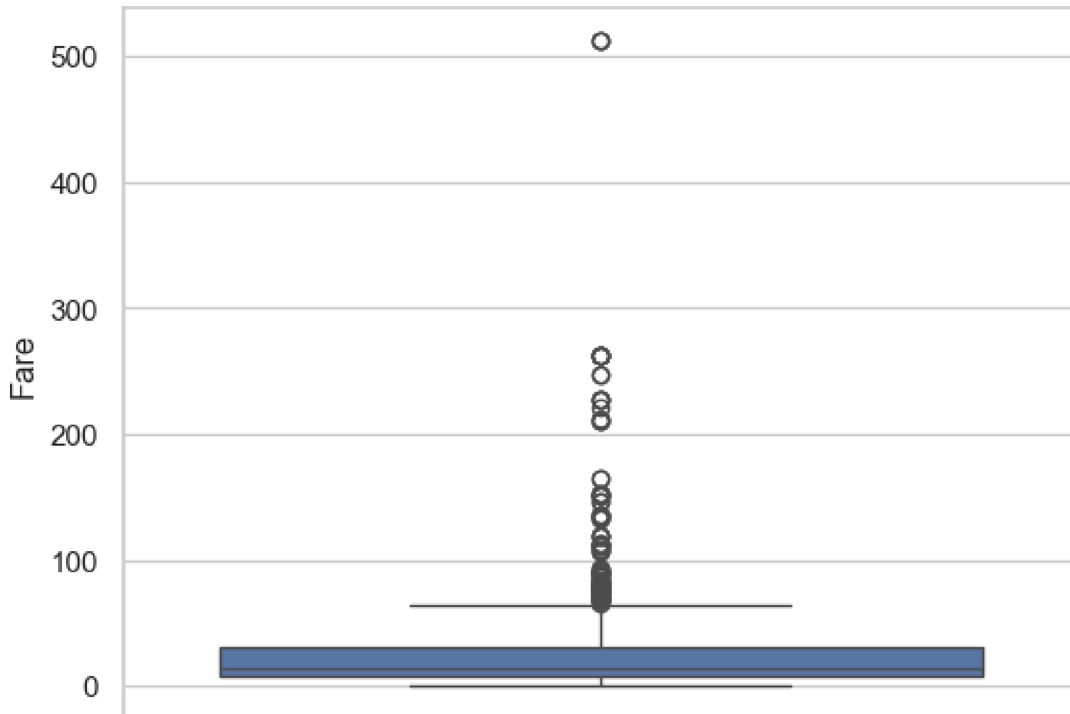
[60]:
```
# Class distribution:
import matplotlib.pyplot as plt
sns.countplot(x='Pclass', data=train_df)
plt.show()
```



[62]:
```
# Age distribution:
import matplotlib.pyplot as plt
sns.histplot(train_df['Age'].dropna(), bins=30)
plt.show()
```

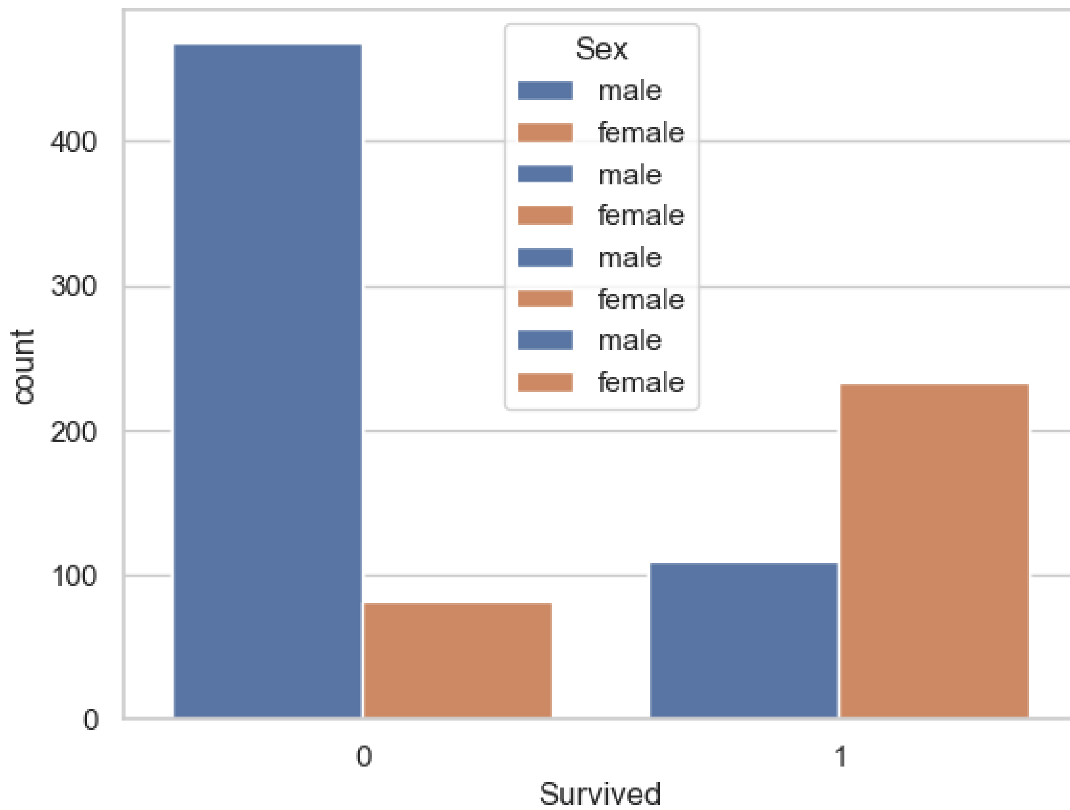


[64]:
```
# Boxplot for Fare:
import matplotlib.pyplot as plt
sns.boxplot(y='Fare', data=train_df)
plt.show()
```

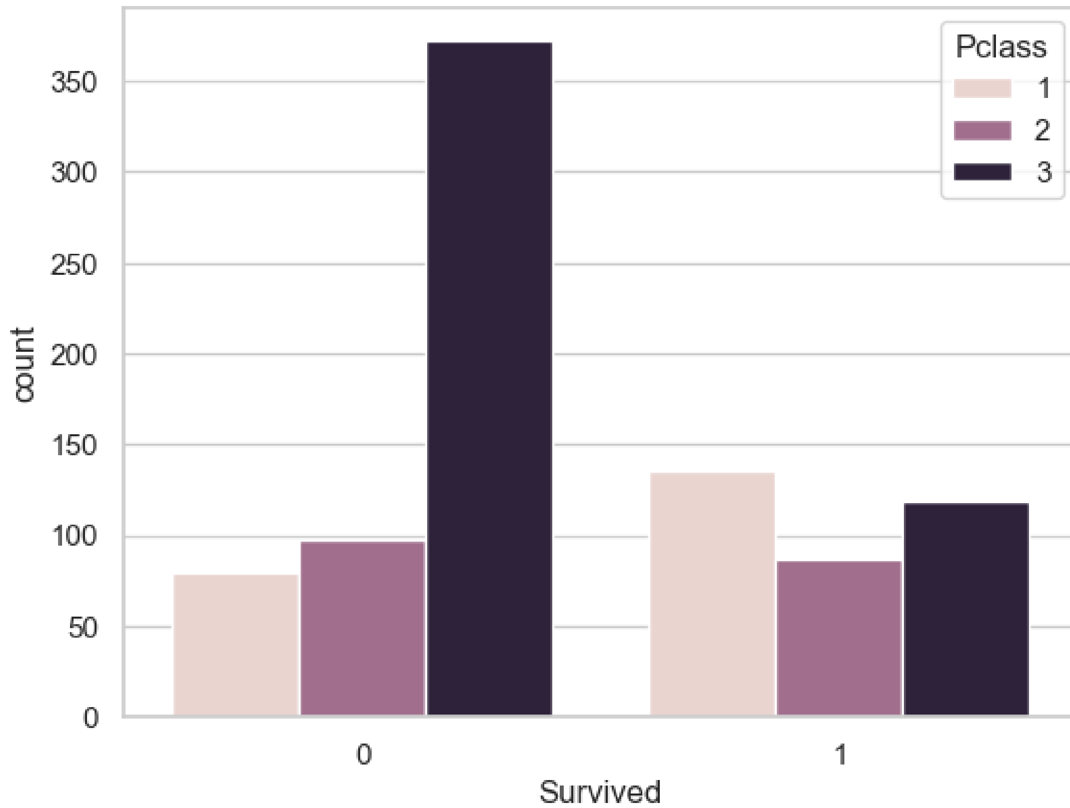


1.0.2 STEP 5: Bivariate Analysis (Two Variables Together)

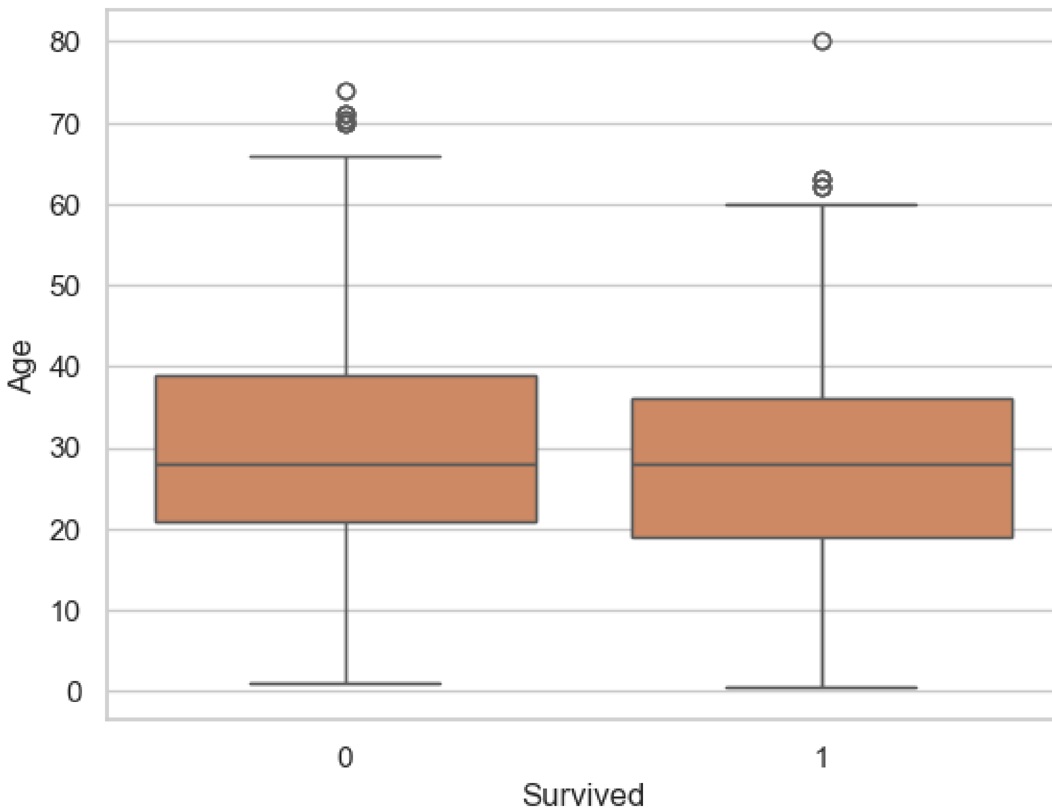
[75]:
```
#Survival by Sex:
import seaborn as sns
import matplotlib.pyplot as plt
sns.countplot(x='Survived', hue='Sex', data=train_df)
plt.show()
```



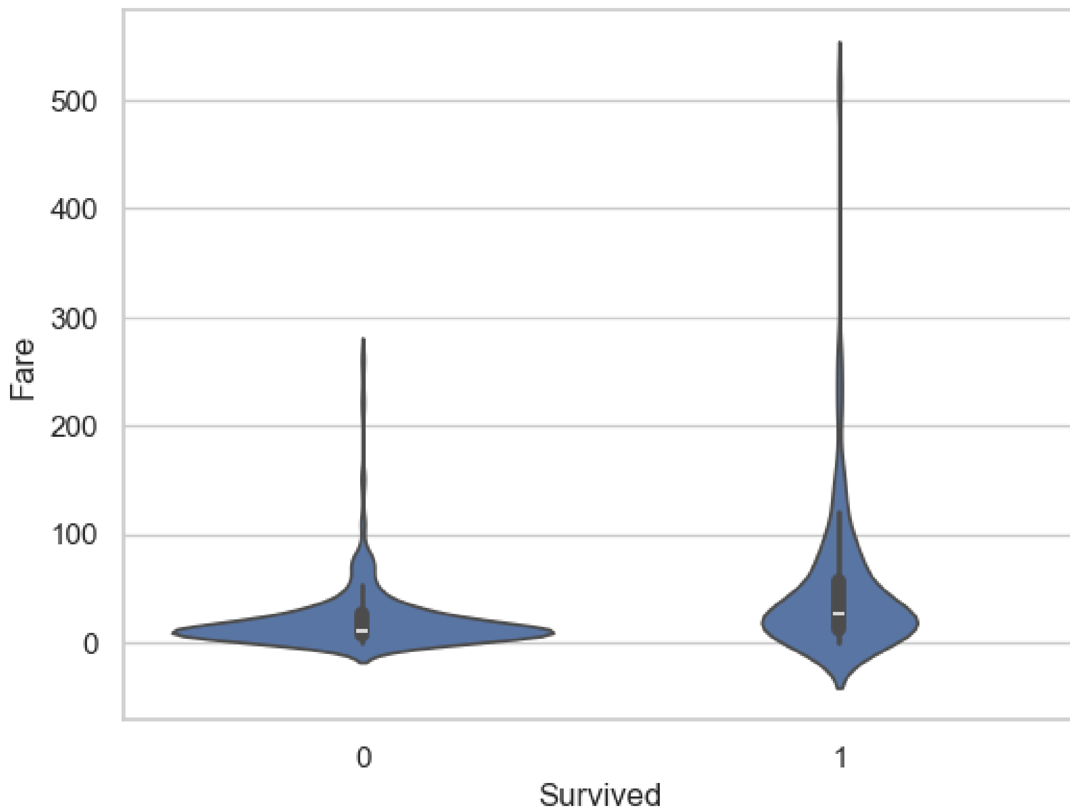
[77]:
```
#Survival by Pclass:
import seaborn as sns
import matplotlib.pyplot as plt
sns.countplot(x='Survived', hue='Pclass', data=train_df)
plt.show()
```



[81]:
```
#Boxplot: Age vs Survived:
import seaborn as sns
import matplotlib.pyplot as plt
sns.boxplot(x='Survived', y='Age', data=train_df)
plt.show()
```

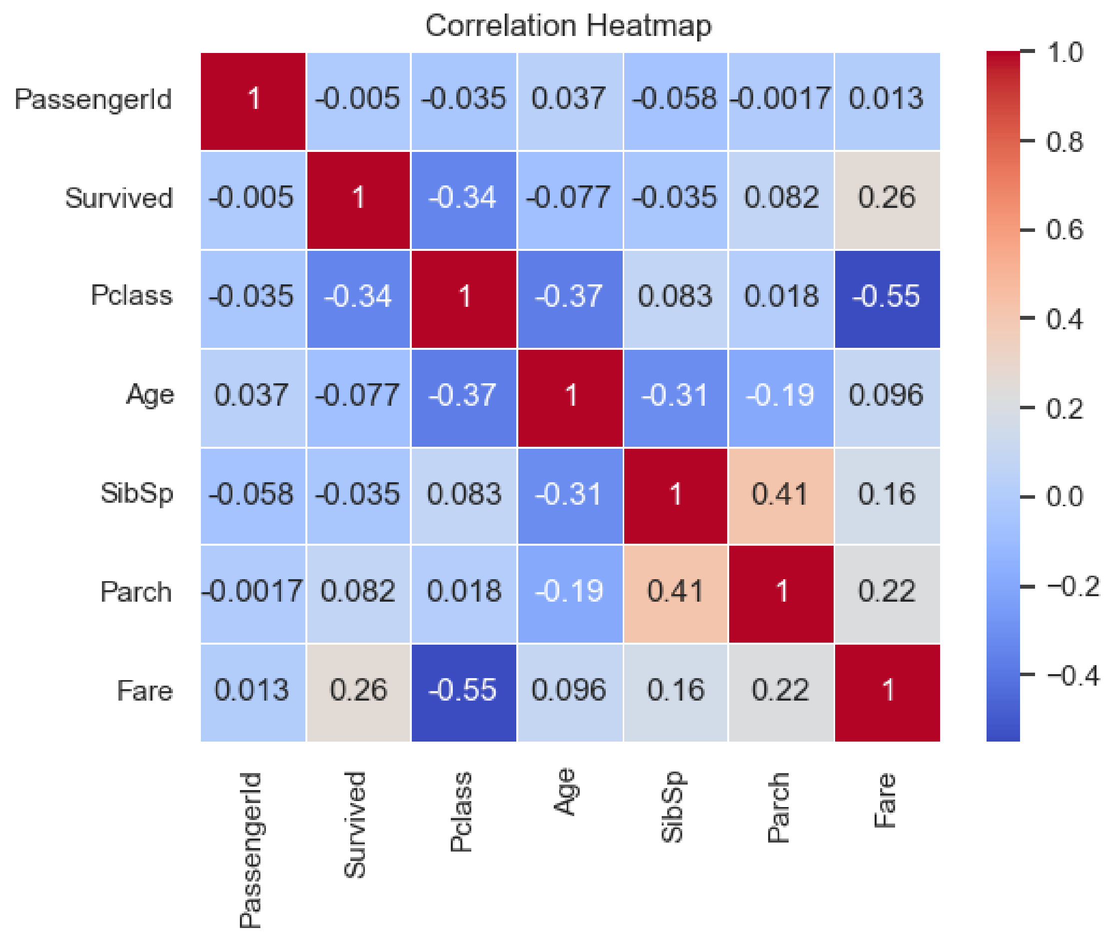


[85]:
```
#Fare vs Survived:
import seaborn as sns
import matplotlib.pyplot as plt
sns.violinplot(x='Survived', y='Fare', data=train_df)
plt.show()
```

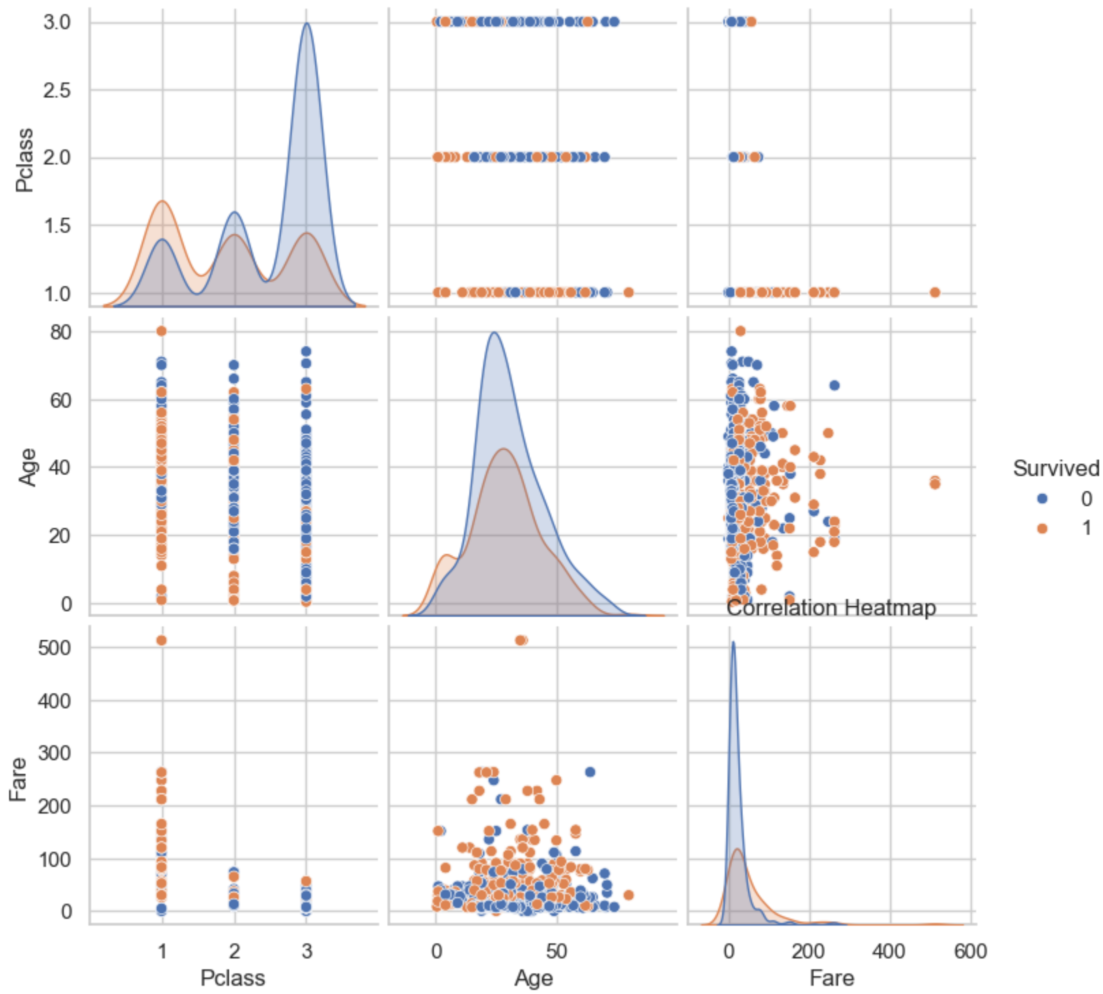


1.0.3 STEP 6: Multivariate Analysis

[92]:
```
import seaborn as sns
import matplotlib.pyplot as plt

# Step 1: Select numeric columns
numeric_df = train_df.select_dtypes(include=['number'])

# Step 2: Correlation matrix
corr_matrix = numeric_df.corr()

# Step 3: Create heatmap
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', linewidths=0.5)

# Step 4: Show plot
plt.title('Correlation Heatmap')
plt.show()
```



[94]:
```
#Pairplot:
import seaborn as sns
import matplotlib.pyplot as plt

# Step 1: Select numeric columns
numeric_df = train_df.select_dtypes(include=['number'])

# Step 2: Correlation matrix
corr_matrix = numeric_df.corr()

# Step 3: Create heatmap
sns.pairplot(train_df[['Survived', 'Pclass', 'Age', 'Fare']].dropna(), hue='Survived')
# Step 4: Show plot
plt.title('Correlation Heatmap')
plt.show()
```



1.0.4 STEP 7: Data Cleaning (Optional for Insight Clarity)

[105]:
```
# Fill missing values for multiple columns with their medians
train_df[['Age', 'Fare']] = train_df[['Age', 'Fare']].fillna(train_df[['Age',␣
↪'Fare']].median())
# Check if missing values are filled
print(train_df[['Age', 'Fare']].isnull().sum())
```

```
Age     0
Fare    0
dtype: int64
```

[117]:
```
# Drop Cabin if too many nulls:
if 'Cabin' in train_df.columns:
    train_df.drop(columns=['Cabin'], inplace=True)
```

Summary: ## Titanic Dataset Visual Insights

1.0.5 Age Distribution

Observation:
The age distribution is right-skewed, indicating a larger number of younger passengers onboard the Titanic, with the majority aged between 20 to 30 years.

1.0.6 Survival by Passenger Class

Observation:
Survival rates varied significantly across passenger classes, with first-class passengers having a notably higher survival rate compared to second and third-class passengers.

1.0.7 Survival by Gender

Observation:
Females had a much higher survival rate than males, indicating that women were given priority during evacuation.

1.0.8 Survival by Embarked Port

Observation:
Passengers embarking from Cherbourg (C) had the highest survival rate, while those from Queenstown (Q) had the lowest, suggesting possible differences in demographics or cabin locations.

1.0.9 Siblings/Spouses Aboard

Observation:
Passengers without siblings/spouses aboard had a lower survival rate compared to those with 1–2 family members, suggesting moderate family presence may have improved chances of survival.

1.0.10 Age and Gender Survival

Observation:
Children (age < 10) had a relatively high survival rate, especially females, whereas male passengers across most age groups had lower survival rates.

1.0.11 Fare vs Class and Survival

Observation:
Most passengers who paid higher fares were in the first class and had better survival rates, indicating that fare price and class were strong indicators of survival probability.

1.0.12 Cabin Info and Survival

Observation:
Cabin information is missing for the majority of passengers, but among the available data, those with cabins had higher survival rates, possibly due to their first-class accommodations.

1.0.13 Family Size and Survival

Observation:
Family size affects survival—small families (1–3 members) had higher survival rates compared to solo travelers or large families, showing a U-shaped relationship.

[ ]:
```

```